Replicate-First: Honest Uncertainty, a Flux Law, and Complete Mechanism Identification for Matched-Pair Growth Studies on Evolving Dependency Graphs

Author. Chris Brink (independent) **Version.** Draft v0.2, 2026-09-02 (v0.2: adds the Phase E registered kill to §4). Assembled from the accretion-study laboratory (`accretion-study/`: `SPEC.md`, `SPEC-D.md`, `THEORY.md`, scripts `01–06` with registered results committed beside them) and the five-generation field program it audited (`debian-study/`, `battery-v2/–battery-v5/`). This paper contains no new runs: every number cites a registration committed before its single scored execution. **Status discipline.** Claims carry grades: **[C]** classical mathematics, proved in text or cited; **[computed]** registered finite computation, single scored run per registration; **[A]** interpretive. The grades are load-bearing.

Abstract

Matched-pair designs are a standard instrument for growth claims on evolving networks: pair structurally similar nodes, difference their future gains, and read significance off a permutation null. We report three results about this instrument, each established in a synthetic laboratory where ground truth is available by construction, and then demonstrated end-to-end on a real registered program whose final act was killing its own headline claim. **(1) The audit.** The matched-pair point estimator is unbiased on generators provably blind to the tested partition — exactly under uniform attachment, by exchangeability under preferential attachment [C] — but the standard within-corpus sign-flip null understates true across-universe variance by roughly an order of magnitude [computed]. The failure has two levels, only one of which any within-corpus resampling can price: pairs sharing a kernel share a neighborhood (repairable by clustered flips), and all pairs share the one realized growth history (not repairable by any rearrangement of it). For a corpus that exists once, ensemble uncertainty is not samplable from within; the honest warrant is replication across independent corpora and sealed out-of-sample prediction. **(2) The flux law.** Under cone-local accretion — newcomers pick a platform and draw dependencies from its truncated transitive closure — expected per-step gain is affine in **ORACLE(x) = Σ_{u : x ∈ cone(u)} 1/|cone(u)|**, harmonic cone-membership mass [C-style derivation, confirmed 10/10 universes]. This quantity is absent from standard predictor batteries (degree, age, PageRank, k-core), which is a constructive proof that a baseline-computable partition can carry growth signal beyond any such battery without magic. **(3) Complete mechanism identification.** In the laboratory, matching on ORACLE extinguishes the partition's signal exactly ($t = -0.25$ from $t = -10.2$), while matching on exact *uncapped* descendant counts does not ($t = -7.0$): the operative structure is concentration of reach, not volume [computed]. The field demonstration: five sealed battery generations on the Debian archive, in which the laboratory's derived features killed two prior certified effects on another corpus, and an institutional-metadata control — outside the graph, hence outside the audit's guarantee — finally killed the program's own four-generation-stable headline. We distill the method into a checklist we commit to and recommend: replicate-first designs for generators, cross-corpus replication and sealed bets for single-history corpora, flux-law features and non-structural covariates in the battery from generation one, and interpretation tables fixed before every run.

1. Setting

A growing directed graph is observed at snapshots; a study fixes a baseline snapshot, computes node features and a node partition from the baseline only, and asks whether partition membership predicts in-degree gain over a horizon beyond the features. The estimator throughout: for a sampled set of *kernels* (reference nodes making the partition non-degenerate), candidate nodes are grouped by partition cell and exact undirected graph distance to the kernel's territory; cross-cell pairs are matched greedily without replacement within a 0.5-caliper ball in the z-space of the feature battery, subject to a pre-registered balance gate (max |SMD| ≤ 0.10 per feature); the statistic is the mean within-pair gain difference; the conventional null flips pair signs. The partition under study is the aperture program's four-cell structure (Exploitation / Refusal / Infrastructure / Distribution; `preprints/aperture/paper.md`), but nothing in this paper depends on its specifics: the results apply to any baseline-computable node classification tested by matched pairs.

The laboratory (`accretion-study/sim-lib.mjs`) grows synthetic universes from fixed rules — uniform attachment (U), preferential attachment (PA), and platform-cone accretion PC(β), where a newcomer picks a platform uniformly and draws dependencies from the platform's truncated down-set (β mixes cone-local with global draws) — and runs the *identical* estimator code on them. Two features of real corpora proved structurally necessary for non-degeneracy and are part of the recipe: root nodes must exist (dependency count $m = 0$ with positive probability; otherwise the first node is a universal ancestor and the Refusal cell is empty everywhere), and platform choice must not be popularity-weighted from birth (or primordial hubs recreate the universal ancestor). Both failures, and their fixes, are documented in `SPEC.md` [computed].

2. The audit: what permutation percentiles measure

Proposition 1 (uniform attachment is exactly null) [C]. Under U, for any pairing rule measurable in the baseline graph, $E[\Delta] = 0$. *Proof:* at each future step every existing node is hit with equal probability regardless of any baseline property; conditional expected gain is a shared constant; pair differences vanish in expectation by the tower property. ∎

Proposition 2 (PA is null given in-degree matching) [C]. Under PA, baseline nodes of equal in-degree have exchangeable futures (the transition law sees only the weights), hence equal expected gains; any exact-in-degree pairing has $E[\Delta] = 0$. ∎ (The estimator matches log in-degree within a caliper, not exactly; the balance gate bounds the residual, and replicate calibration measured means indistinguishable from zero on both U and PA, as the propositions predict [computed].)

The variance failure [computed]. Replicate calibration — many independent universes per rule, the estimator run once per universe — found the across-universe standard deviation of the statistic roughly **10× the sign-flip band**, on generators where the truth is null. The percentile attached to any single universe's run is therefore not a generator-level tail probability; confident false positives on provably null rules are routine. Kernel-clustered flips (flipping each kernel's pairs as a block) were tested and found insufficient (`01c-validate-clustered-null.mjs`).

Why, precisely. $\mathrm{Var}(\mathrm{mean}) = \Sigma d^2/n^2$ is correct only if pair differences are uncorrelated given magnitudes. They are correlated at two levels: *within kernel* — pairs sharing a kernel share the local neighborhood, and a growth wave moves them together (clustered flips price exactly this) — and *across the universe* — which regions of the graph are fertile over a horizon is a property of the one realized trajectory, and every kernel rides it. No within-corpus rearrangement can price the second level, because every rearrangement sees the same single trajectory. A within-corpus percentile is a statement about pair-exchangeable rearrangements of one history, not about the ensemble of histories.

Consequences for practice. For synthetic generators, run replicate-first: the universe, not the pair, is the unit of inference (the laboratory's confirmatory phase used 20 universes per rule and t-statistics across universes [computed]). For a real corpus — which exists once — the ensemble is not samplable from within; report percentiles only as conditional statements, and carry the warrant with replication across independent corpora and sealed out-of-sample directional bets registered before first measurement. We adopted this reading discipline retroactively for an entire field program the same day the calibration landed, repricing every previously reported percentile (`SPEC.md` postscript; the briefs' dated corrections).

3. The flux law, and the feature batteries cannot see

Under PC(0), fix a node x and let up(x) = {u : x ∈ cone(u)} be the set of nodes whose truncated dependency cone contains x. Per growth step,

$P(x \text{ gains}) \approx (1/t) \cdot [\ 1 + (\bar{m} - 1) \cdot \Sigma_{\{u \in up(x)\}}\ 1/|cone(u)|\ ]$,

one term for being drawn as platform, one for being sampled from a cone containing x. Expected gain is affine in **harmonic cone-membership mass**, ORACLE(x) = $\Sigma_{\{u \in up(x)\}}\ 1/|cone(u)|$: *being a large share of many small toolchains* [C-style derivation, `THEORY.md` §3; direct confirmation, 10/10 universes, `03-sign.mjs`].

Standard batteries (in/out-degree, age, PageRank, k-core) contain no up-set measure — PageRank in the depended-upon orientation is a damped, normalized cousin, truncated hard by both. Consequently a generator whose flux tracks cone mass leaves room for a baseline-computable partition to carry growth signal *at matched battery*. PC(0) realizes this: its cell ordering R > E > D holds at t ≈ 10 across 20 universes against the full five-feature battery [computed]. This is a constructive possibility proof with two edges, and we state both: a partition can legitimately out-inform a reviewer's arsenal (no leakage, no artifact — the battery is simply blind to the operative quantity), and *any* field claim certified "beyond battery X" is one derived feature away from reclassification.

4. Complete mechanism identification, demonstrated

The laboratory then closed its own loop under registration (`05-oracle.mjs`) [computed]:

- **O1.** ORACLE out-predicts truncated descendant *count* in 10/10 universes (r ≈ 0.44 vs 0.23).
- **O2.** With ORACLE added to the battery, the PC(0) cell effect vanishes exactly: Δ_{ER} = −0.004, t = −0.25 (from t = −10.2). The mechanism is *completely* identified — nothing residual remains for the partition to explain.
- **O3.** With exact **uncapped** descendant counts matched instead, the effect persists at t = −7.0. Volume of reach is not the structure; **concentration** of reach is, and only the harmonic weighting expresses it.

We know of no published matched-pair growth study that identifies the complete mechanism of its own effect in closed form and then demonstrates extinction-by-matching. The loop is available to any study with a generative model of its domain: derive what the model rewards, express it as a baseline feature, and aim it at your own result.

A registered kill from the same loop (Phase E) [computed]. ORACLE, repurposed as a descriptive ranking on real corpora, surfaces a distinctive head signature: packages whose concentration rank far outruns their dependent-count rank. We registered the conjecture that this signature is a mechanism fingerprint — present in real corpora and in cone-local generators, absent under the null rules — with statistic, thresholds, and an interpretation table fixed before one run (`06-signature.mjs`). The real half landed (all three corpora, 20–33 of the top 50). The synthetic half failed in both directions at once: uniform attachment *shows* the signature (mean 16 of 50), and the cone-local family — the very mechanism whose growth signal ORACLE completely explains — produces *exactly zero*, in every replicate. Per the pre-fixed table, the conjectured use died on the spot. The inversion it leaves behind is clean: cone-local draws weld concentration rank to volume rank at the head, because within-cone sampling keeps handing direct dependents to exactly the nodes sitting in many cones; global draws let deep old nodes accumulate cone membership without proportional in-degree, opening the gap. Divergence between concentration and volume is a signature of growth where newcomers can attach anywhere — not of the mechanism ORACLE was derived from. We include this kill because it is the checklist of §6 operating at full speed: conjecture to registered run to verdict in one afternoon, thresholds untouched after the result.

5. The field demonstration: five batteries, four kills

The method's field trial was a registered program on real dependency corpora, run at full adversarial volume (primary sources: `debian-study/`, `battery-v2/`–`battery-v5/`; narrative account: `preprints/seedbed/paper.md`). In brief [all computed]:

Generation	Knife added	Origin	Outcome
v1	fine-grain degree, out, age, PageRank, k-core	reviewer's arsenal	killed crates.io (+15.7 → popularity); Debian sealed bet landed (+0.098)
v2	truncated up-set size	flux law (§3)	killed Go E-over-R (+0.22 → up-set imbalance, unrepairable); Debian held
v3	ORACLE	mechanism (§4)	killed Go E-over-D (+0.152 → +0.018); Debian held
v4	momentum (prior-interval gain)	standard autoregressive control	Debian held (+0.083; marginal contribution +0.002)
v5	functional role (archive Section, exact stratum)	institutional metadata	**killed Debian** (+0.083 → −0.076; within-role null-to-negative, reversal javascript-dominated)

Two lessons carry beyond the program. First, the theory-derived knives (v2, v3) killed effects that had survived everything a reviewer would ask for — derived confounds are sharper than guessed ones. Second, the fatal covariate for the final claim was *not a function of the graph*: package role lives in archive metadata, and the audit of §2 — which certifies the estimator against structural confounds on generators — is constitutionally silent about covariates outside the object the generator produces. A matched-pair estimate stable under any number of structural controls is evidence only about the controls you have. The program's blind pre-check protocol surfaced the danger before the run (role imbalance leaned *with* the deflating story, the first knife to do so), the registration fixed the kill condition before the verdict, and the claim's obituary was published at the volume its survival would have received, the same day.

6. The checklist

For any matched-pair growth study on an evolving network, we commit to and recommend:

1. **Replicate-first for anything with a generator.** The universe is the unit of inference. Single-universe percentiles on synthetic data are not evidence.
2. **For single-history corpora:** percentiles as conditional statements only; warrant from independent-corpus replication and sealed directional bets registered before first measurement.
3. **Battery contents, from generation one:** the standard arsenal, *plus* truncated up-set size, *plus* ORACLE (both pure graph features, cheap at cap 200), *plus* the node's own prior-interval gain, *plus* every available non-structural covariate of node kind (role taxonomies, categories, institutional metadata) — the last as exact strata, not z-features, where cardinality permits.
4. **Blind pre-checks** (balance and composition audits touching no outcomes) before every registered run; publish them with the run.
5. **Interpretation tables fixed before execution**, including what kills the claim; single scored run per registration; kills reported at the volume of wins.
6. **If a generative model of the domain exists, derive its reward functional and aim it at your own result** before anyone else does.

7. Reproducibility

Every number above cites a registration committed before its single scored run; seeds are constants in the scripts; raw outputs are committed beside them. Laboratory chain: `accretion-study/SPEC.md` → `01-explore.mjs` → `01b-calibrate.mjs` → `01c-validate-clustered-null.mjs` → `02-confirm.mjs` → `THEORY.md` → `SPEC-D.md` → `03-sign.mjs` → `04-explore-d.mjs` → `05-oracle.mjs` → `06-signature.mjs`. Field chain: see `preprints/seedbed/paper.md` §8. Repository: github.com/thefalsework/papers. The ORACLE ranking tool referenced in §4 Phase E is published separately as conemass (github.com/thefalsework/conemass), with dated rankings.

Disclosure. Drafting was AI-assisted under direction; the grades are the author's warrant.